~~PY21IAT~~ / 22PH12B / 22PH22B
PY21ITB

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

I / II Semester B. E. Regular / Supplementary Examinations Feb -2024

Common to AS / CH / IM / ME

CLASSICAL PHYSICS FOR ENGINEERS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer SIX full questions from Part B. In Part B question number 2 and 11 are compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8 & 9 and 10.
3. Handbook of Physics is allowed.

PART-A

1	1.1	Define Poisson's ratio.	01
	1.2	What is the strain of an incompressible body?	01
	1.3	In spite of having shock absorber, why drivers go slowly over bumps?	01
	1.4	What is the damping term in mechanical oscillators?	01
	1.5	What is the solid to gas phase transition process called?	01
	1.6	What is irreversible process in Thermodynamics?	01
	1.7	Give an example of capillary motion.	01
	1.8	What is viscosity of a fluid?	01
	1.9	What are photoelectrons?	01
	1.10	Define Tensile strength of a material.	01

PART-B

2	a	With a neat labeled diagram, setup and solve the differential equation governing the dynamics of an under-damped oscillator. Represent the solution graphically.	10
	b	A $2D$ square plate of mass $1kg$ and length $10cm$ is suspended by a wire which passes through its face center and perpendicular to its plane. Find the time period of angular oscillation for small displacement if the torque required per unit twist of the wire is $4.5 \times 10^{-3}\ N-m/radian$.	04
3	a	With a neat diagram, derive an expression for the couple per unit twist of a solid shaft of radius R and length L. What would be the difference if we take a hollow shaft instead, with inner radius R_1 and outer radius R_2?	10
	b	Calculate the sharing modulus of a body which has a bulk modulus of $8.0 \times 10^{10} N/m^2$ and Poisson's ratio of 0.25.	04
		OR	
4	a	Define briefly any 2 types of stress and strain. Show that a shearing stress is equal to a tensile stress and a compressive stress.	10
	b	A load of $5kg$ produces an extension of $1mm$ in a wire of $5m$ in length and $1mm$ in diameter. Calculate the Young's modulus of the wire.	04

	a	Derive an expression for the work done in a adiabatic process, after deriving the adiabatic gas law equation. State 2 limitations of the 1st law of thermodynamics.	10
	b	When the valve of a completely evacuated bottle is opened, $0.2m^3$ of the atmospheric air at a pressure of $101.325kPa$ rushes in. Calculate the work done by air.	04
		OR	
6	a	With a diagram explain Joule's experiment demonstrating the equivalence of heat and work. Show how it leads to the concept of internal energy, hence explain the first law of thermodynamics for a cyclic process.	10
	b	A stationary mass of gas is compressed isobarically ($P = 200kPa$) without friction from $0.3m^3$ to $0.15m^3$. Calculate the change in internal energy if there is a transfer of $38.7kJ$ of heat from the gas during the process.	04
7	a	Diagrammatically explain the difference between stream line, streak line and path line. Derive the continuity equation in Integral form.	10
	b	A stream of water emerging from a faucet of area $1.2cm^2$, falls through a vertical distance of $85mm$ under gravity whereby its area become $0.12cm^2$. Calculate the rate at which water flows from the faucet.	04
		OR	
8	a	Discuss briefly the construction, working principle and procedure of an ultrasonic interferometer.	10
	b	A $1mm$ diameter glass tube is inserted into water at 20^0C in a cup. Determine the capillary rise of water in the tube. Given the surface tension of water at 20^0C is $0.073N/m$. Given density of water is $1000kg/m^3$. Assume contact angle for water is 10^0.	04
9	a	Describe in details the optical and $I-V$ characteristics techniques of a material.	10
	b	What is yield strength of material? What method is used to test it?	04
		OR	
10	a	Explain the principle, construction and working of an atomic force microscope with neat diagram. Mention two of its applications.	10
	b	Discuss the basic working principle and use of a X ray diffractometer.	04
		LAB COMPONENT	
11	a	In an experiment on torsional pendulum, the following data was obtained	

Configuration	Time for 10 oscillations	
Circular disc ($R = 7.5\ cm$), Mass = $1.1\ kg$ Wire passing through centre of gravity And perpendicular to its plane	i) 39.1 s ii) 40.2 s iii) 38.9 s	39.4
Rectangular disc (length $15cm$, breadth $7cm$) wire is attached perpendicular to both length and breadth mass = $1.2\ kg$	i) 24.5 s ii) 24.2 s iii) 24.1 s	24.26

$1.93 \times 10^{-4} = I_1$

$= 0.199 \times 10^{-3}$

$4 \times 10^{-4} = I_2$

$5 \times 10^{-3} = \frac{I_2}{T_2^2}$

b	Calculate the rigidity modulus of the wire, given the length of the wire is $20cm$ and radius of the wire is $0.3mm$ By constructing a ray diagram, explain the apparatus required and procedure involved to find the wavelength of a laser beam by diffraction technique and calculate the wavelength of a laser beam from the data given below. Given the grating constant is $5.08 \times 10^{-5}m$ and the distance between grating and screen is $100cm$.	10

Diffraction order (n)	*Distance* $2x_n$
1	2.6
2	5.4
3	8.2
4	10.8
5	13.8
6	16.4

10

Scheme and solution for 22PHY12 BAE Code: N22PHY12BAE Feb 2024 CP for E

PART-A

1.1) $\sigma = \frac{\Delta r/r}{\Delta L/L}$ — 1m

1.2) Incompressible body ⇒ Strain = 0 — 1m

1.3) The spring in the shock absorber is made to vibrate in the elastic range by going slowly — 1m

1.4) Friction is the damping term in mechanical oscillation — 1m

1.5) Sublimation

1.6) ~~Due to friction the body has to spend more~~ system and ~~the surroundings~~ do not return to their original condition after a cycle is completed. This is an irreversible process.

1.7) Water rising in a pippete against gravity is an example of capillary motion — 1m

1.8) Internal resistance developed as a fluid to it's flow is called Viscosity — 1m

1.9) Electrons emitted from the surface of a material with low work-function when incident upon by photons of high energy are called photo-electrons. — 1m

1.10) Tensile strength is the maximum stress that a material can bear before it breaks. — 1m

PART-B

2a) Diagram of damped oscillator — 1 mark

Differential Eqn: $m\ddot{x} + r\dot{x} + kx = 0$ — 1mark

$\Rightarrow \ddot{x} + \frac{r}{m}\dot{x} + \frac{k}{m}x = 0 \Rightarrow \ddot{x} + 2b\dot{x} + \omega^2 x = 0$ — 2marks

$x = Ae^{\alpha t}$ (Trial soln) — 1mark

$\Rightarrow (\alpha^2 + 2b\alpha + \omega^2) = 0 \Rightarrow \alpha = -b \pm \sqrt{b^2 - \omega^2}$ — 1 mark

Underdamped ⇒ $b < \omega \Rightarrow \omega^2 - b^2 = \beta^2 \Rightarrow \alpha = -b \pm i\beta$ — 1mark

$\Rightarrow x = Ae^{-bt}\sin(\omega' t + \phi)$ — 1mark

Graphical representation — 1mark

For worst small damping $\omega' = \sqrt{\omega^2 - b^2}$ — 1mark

scheme and solutions for ...

b) Square plate, axis through face centre $\Rightarrow I = \frac{M}{12} \times 2L^2 = \frac{M}{6}L^2$ — 1mark

$C = 4.5 \times 10^{-3}$ N-m/radian — 1m

$T = 2\pi\sqrt{\frac{I}{C}} = 2\pi\sqrt{\frac{ML^2}{6C}}$ — 1m

$T = 2\pi\sqrt{\frac{(.1)^2}{6 \times 4.5 \times 10^{-3}}} = 2\pi\sqrt{\frac{10}{27}} = \frac{2\pi}{\sqrt{2.7}} = 3.82\,s$ — 1mark.

3a) Diagram — 1mark, Strain, Stress, Force, Moment — 4 marks,

Integrate moment to get Couple per unit twist: $C = \frac{\pi\eta R^4}{2L}$ — 2 mark

For hollow shaft, integrate $\int_{R_1}^{R_2} dM = \frac{\pi\eta(R_2^4 - R_1^4)}{2}$ — 2 mark

3b) $\eta = \frac{Y}{2(1+\sigma)}$, $K = \frac{Y}{3(1-2\sigma)} \Rightarrow Y = 3K(1-2\sigma)$

$\Rightarrow \eta = \frac{3K(1-2\sigma)}{2(1+\sigma)}$ — 2marks $\Rightarrow \eta = \frac{3}{2}\frac{(1-0.5)}{1.25} \times 8 \times 10^{10}$ — 1 mark

$\eta = \frac{1.5 \times 0.5 \times 8}{1.25} \times 10^{10} = 4.8 \times 10^{10}\ N/m^2$ — 1mark

4a) Any 2 types of Stress and Strain definition — 4 × 1mark

Unit cube diagram — 2 marks, Tensile Stress — 1 mark,

Compressive Stress — 1mark, Final statement — 2marks

4b) $Y = \frac{F/A}{\Delta L/L} = \frac{FL}{A\Delta L}$ — 1mark $= \frac{4mgL}{\pi d^2 \Delta L}$ — 1mark

$= \frac{4 \times 5 \times 9.8 \times 5}{3.14 \times 10^{-6} \times 10^{-3}} = \frac{980}{3.14 \times 10^{-9}} = \frac{98}{3.14} \times 10^{10} = 31.2 \times 10^{10}\ N/m^2$

5a) Derive Gas Law Eqn (PV^γ = Constant), taking $\Delta Q = 0$ — 5m

$W = \int dW = K\int V^{-\gamma} dV = \frac{K}{1-\gamma} V^{1-\gamma}$ — 1 marks

Simplifying we get final answer $W = \frac{P_i V_i - P_f V_f}{\gamma - 1}$ — 3marks

2 limitations of 1st Law are:-

i) Cannot comment about the feasibility/probability of a particular thermodynamic process

ii) Cannot predict if a process is spontaneous or not. } 2 marks

5) Since vacuum-air boundary is decreasing, work done by air is -ve — 1 mark.

Bottle is incompressible $\Rightarrow W_{bottle} = 0$ — 1 mark

So work done by air $= \int p\,dV = p\,\Delta V = 101.325 \times 10^3 \times 0.2$

$= 20.265$ kJ — 1 mark.

Work done by air $= -20.265$ kJ — 1 mark

6a) Joule's experiment. diagram — 2 marks, Description — 3 marks

Internal Energy definition — 1 mark. Show how concept evolved from Joule's experiment — 1 mark.

Using concept of U, write the 1st law of Thermodynamics — 1 mark

Explain the meaning of the equation — 2 marks

6b) $\Delta Q = \Delta U + \Delta W \Rightarrow Q_{12} = (U_2 - U_1) + W_{1-2}$ — 1 mark

$W_{12} = -P(V_2 - V_1) = -200 \times 10^3 (0.3 - 0.15) = -3 \times 10^4$ J — 1 mark
$= -30$ kJ

$Q_{12} = -38.7$ kJ — 1 mark

So $-38.7 = (U_2 - U_1) - 30$ kJ

$\Rightarrow U_2 - U_1 = -38.7 + 30 = -8.7$ kJ — 1 mark

7a) Stream Line, Streak Line, Path Line, draw diagram and explain — 3 x 2 marks

Continuity eqn in integral form — 4 marks

7b) From continuity eqn, we get $A_1 V_1 = A_2 V_2$ — 1 mark

Faucet A_i, V_i $\quad V_f^2 = V_i^2 + 2gh$ — 1 mark

A_f, V_f $\quad \Rightarrow V_f$ (Rate of water flow from faucet)

$= \sqrt{\dfrac{2gh\,A_f^2}{A_i^2 - A_f^2}}$ — 1 mark

$= \sqrt{\dfrac{2 \times 9.8 \times .085 \times (0.12)^2 \times 10^{-4}}{\{(1.2)^2 - (0.12)^2\} \times 10^{-4}}} = \sqrt{\dfrac{0.02399}{1.4256}} = 0.129$

$= 12.9$ cm^3/s. — 1 mark

Scheme and Solution for 22PHY12D

8a) Ultrasonic Interferometer: Diagram – 3 marks, Construction – 2 marks
Working principle – 3 marks, Procedure – 2 marks.

8b) $h = \frac{2\sigma_s}{\rho g R}\cos\phi = \frac{2 \times 0.073}{1000 \times 9.8 \times 0.5 \times 10^{-3}}\cos(10^\circ)$ – 2 marks

$= \frac{0.14378}{4.9} = 0.02934 = 2.9$ cm. – 2 marks

9a) Optical characterization – Method and diagram – 3 + 2 (ellipsometry)
I – V characterization – Method and diagram – 3 + 2 (Solar cell)

9b) Yield strength definition – 2 marks
Max stress before plastic deformation begins.
UTM machine is used to test yield strength. } 2 marks
Explain UTM in brief –

10a) Diagram of AFM – 2 marks
Principle " " – 2 marks
Construction – 2 marks
Working – 2 marks
} 8 marks

2 Applications of AFM: a) Test surface smoothness, b) Test surface potential and interactions – 2 marks

10b → XRD – Bragg's law (3) + Uses (1).

11a) Torsional pendulum, calculate $\frac{I}{T^2}$ for circular disc

using $I_{cir} = \frac{MR^2}{2}$

For rectangular disc, $I_{rect} = \frac{M}{12}(L^2 + B^2)$

$\Rightarrow \frac{I_{cir}}{T_{cir}^2} = \frac{I_{rect}}{T_{rect}^2} \Rightarrow \frac{(L^2+B^2)}{12\,T_{rect}^2} = \frac{R^2}{2\,T_{cir}^2}$

$\Rightarrow L^2 + B^2 = 6R^2\left(\frac{T_{rect}}{T_{cir}}\right)^2 \Rightarrow B = \sqrt{6R^2\left(\frac{T_{rect}}{T_{cir}}\right)^2 - L^2}$ – 4 marks

Calculation of T_{cir} – 3 marks, Calculation of T_{rect} – 3 marks

Scheme and solution for ... Oct 2023

Diagram – 1
Apparatus – 1
Procedure – 2

11b) Calculate x_n — 2 mark (1)

$\tan\theta_n = \frac{x_n}{d}$ — 2 marks (2)

$\theta_n = \tan^{-1}\left(\frac{x_n}{d}\right)$ — 2 marks (1)

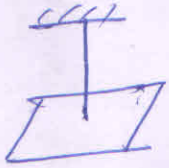
$\sin\theta_n$ — 2 marks, λ — 2 marks (1)

11a)



$\frac{MR^2}{2} = 30.93\times10^{-4}\ kgm^2$ (4)

T (s)	$\frac{I}{T^2}$ ($kgms^{-2}$)
$\frac{39.4}{10}$	9.99×10^{-4} (2)



$\frac{M(L^2+B^2)}{12} = 27.4\times10^{-4}\ kgm^2$

$\frac{24.26}{10}$	9.65×10^{-4}
Avg $\frac{I}{T^2}$	$9.832\times10^{-4}\ kg\ ms^{-2}$

$\eta = \frac{8\pi L}{r^4}\left(\frac{I}{T^2}\right)_{avg} = \frac{8\times3.14\times20\times10^{-2}m}{(0.3\times10^{-3}m)^4}\times0.332\times10^{-3}\ kgms^{-2}$ (4)

$= 20.59\times10^{10}\ Nm^{-2}$

28.02.24

11b). $d = 100cm$, $C = 5.08\times10^{-5}m$

n	$2x_n$ (cm)	x_n (cm)	$\tan(\theta) = x_n/d$	$\theta_n = \tan^{-1}\left(\frac{x_n}{d}\right)$ (in deg)	$\lambda = \frac{C\sin\theta_n}{n}$ (nm)
1	2.6	1.3	0.013	0.745	660.34
2	5.4	2.7	0.027	1.547	685.55
3	8.2	4.1	0.041	2.349	693.68
4	10.8	5.4	0.054	3.0925	684.80
5	13.8	6.9	0.069	3.949	699.38
6	16.4	8.2	0.082	4.690	691.94

Avg: 685.95 nm